Frozen Encoders for Few-Shot Rare Retinal OCT Classification: A Proposed Evaluation Protocol

Doan Tung Lam, Ha Quang Minh, Pham Duc Anh, Ngo Hoang Phuc
National Economics University

***Abstract*—Rare retinal diseases have few labelled optical coherence tomography (OCT) images, so a classifier must learn each disease from a handful of examples. Our earlier CycleGAN-augmentation reproduction showed two problems: the best augmentation changed with the seed, and disease was confounded with source dataset. We propose a test of three frozen foundation models (FMs)—RETFound Nature OCT, DINOv2-Base and MIRAGE-Base—against a matched Keras Inception-v3 reference on five rare diseases. Encoder weights stay fixed. All models share the same labelled real examples (supports) and are tested on held-out images (queries); a cosine prototype is the main classifier and logistic regression with fixed regularization is secondary. Five grouped development folds and paired support draws nominate one FM for reporting before any new test result, and later results cannot change that choice. A separate nine-class audit counts how often NORMAL scans from Kermany, OCTID and OCTDL are labelled as rare, if access and eligibility checks pass. No new FM results are reported. The contribution is a fixed, checkable protocol that states what it cannot show: patient independence, freedom from pretraining exposure and clinical value.**

***Index Terms*—optical coherence tomography, rare retinal disease, few-shot learning, frozen encoders, source shift**

I. Introduction and Problem Statement

OCT gives cross-sectional images of the retina and is widely used to classify retinal disease. With many labelled images, transfer learning recognizes common OCT categories well [1]; Yoo et al. [2] had far fewer images for five rare diseases. We ask two questions. Can a pretrained encoder tell these five diseases apart from a few labelled examples? And does a classifier that also knows the common classes mistake normal scans from a new source for rare disease? Each needs its own test.

Our earlier reproduction of Yoo et al. used four common classes—NORMAL, choroidal neovascularization (CNV), diabetic macular edema (DME) and DRUSEN—and the five rare classes. Frozen Inception-v3 features fed a softmax head trained on one of three rare-class pools: real images only, basic augmentation or CycleGAN images. Table I lists archived accuracies, which are not comparable K-shot results. The best pool depended on the seed: CycleGAN was best with seed 42, while basic augmentation had the highest five-seed mean. The reproduction therefore does not show that GAN images beat basic augmentation; it motivates a paired comparison.

The reproduction also had two validity problems. First, frozen features separated rare from common images almost perfectly: the rare-versus-common area under the curve (AUC) was about 0.995 before and 0.991 after normalizing file format and image framing. Disease and source were confounded, so the audit could not tell whether this separation reflects disease or source. Second, the fixed Kermany test set contains patients who are also in the fitting split.[1] Grouping cannot remove this overlap without changing the legacy benchmark. We therefore split the question in two. A five-class rare-only task removes the common-versus-rare decision. A separate source audit checks whether normal scans receive rare labels. Neither removes acquisition cues within the rare classes.

TABLE I
Historical project accuracy (%); nine-way, fixed legacy test. No new FM scores.

Rare training pool	Seed 42	Seeds 1–5, mean ± SD
Real only	90.31	90.69 ± 1.86
Basic augmentation	88.85	92.93 ± 1.85
CycleGAN augmentation	92.95	92.59 ± 1.29

Sources: archived `results/stage4/metrics.csv` and `results/seed_check.csv`; SD across the five listed training seeds. The same test images were reused.

Rather than proposing a new encoder or adaptation method, we test fixed checkpoints under matched label budgets and audit source-dependent errors. This is also not the first use of a foundation model for rare eye disease. We contribute: (1) a frozen-encoder comparison in which every model sees the same real supports; (2) a development-only rule that nominates one FM before any test outcome; (3) a nine-class audit of false rare predictions on NORMAL scans from three sources; and (4) stop rules that end an arm (one model's pipeline) or the selection, instead of adjusting it, when a check fails. Protocol v1.0 (finalized 5 November 2026) and a claim ledger accompany this paper. All extraction, grouping, selection and inference below are planned.

II. Related Work

A. Rare Retinal Recognition and Pretrained Representations

Yoo et al. [2] added per-disease CycleGAN images to rare OCT classes before nine-class classification. Quellec et al. [3] detected rare fundus diseases with features learned from frequent ones. Both are rare-retina few-shot precedents, but fundus photographs are not OCT, and augmenting from the whole training inventory is not the same as fitting on only K real supports per class.

[1] Archived evidence: `results/c0/v1_gate_summary.csv` and `docs/research/CONTEXT.md`; exact quantities and provenance are mapped in the accompanying claim ledger.

RETFound learns retinal image features and reports how disease detection changes with fewer labels [4]. MIRAGE pretrains on OCT and scanning-laser ophthalmoscopy and benchmarks several OCT tasks [5]. EyeCLIP adds visual-language training and few- and zero-shot ophthalmic tests [6]. DINOv2 is a general self-supervised visual model [7], and generalist-versus-specialist comparisons on eye data already exist [8]. These studies explain our choice of checkpoints. They used different adaptation methods, cohorts and model sizes, so they do not predict which model will win our frozen-cosine task. RETFound's documented Kermany-related pretraining exposure also means the legacy comparison does not control for upstream exposure.

EyeDiff already combines synthetic eye images with foundation models for rare disease [9], so we do not claim that combination as new. Our generators and their automatic feature-distance quality control saw the whole rare training inventory; using fewer of their outputs cannot undo that. We therefore exclude synthetic images from the K-shot comparison.

B. Lessons from Few-Shot Learning Across Fields

Prototypical Networks learn an embedding in which class prototypes classify new examples [10]. SimpleShot shows that simple feature transforms with nearest-neighbor rules are strong baselines [11]. Our classifier differs: it uses frozen pretrained encoders and normalized cosine centroids, so it is neither a newly trained Prototypical Network nor an exact SimpleShot implementation. A cross-domain benchmark covering satellite, dermatology and X-ray images [12] warns that rankings from natural-image benchmarks may not carry over to OCT.

Comparisons must also count resources. STARTUP adapts features with unlabelled target-domain images [13]; we do not allow that extra information. In natural language processing, Mosbach et al. [14] show that adaptation methods must be compared at equal model and example budgets. CLAP's study of vision-language adaptation points to target-specific tuning and sensitivity to distribution shift [15]; its text-initialized, constrained probe is not our fixed-L2 logistic probe. These lessons support matched episodes and open reporting of development labels, not a general claim that linear probes win.

Prompt methods change both the compute and the supervision. CoCoOp learns text prompts conditioned on each image [16]; PathPT combines text prompt tuning with spatial aggregation and tile labels for rare-cancer pathology [17]. Neither is a drop-in image-only OCT baseline. BenchReAD already benchmarks retinal anomaly detection across datasets [18]. Detecting anomalies, however, is not the same as naming five diseases or counting NORMAL scans given a rare label. We keep the encoders frozen and measure their own costs; prompts, adapters and full fine-tuning are out of scope.

III. Proposed Methods

A. Tasks, Data and Information Budgets

The main task is to assign each rare OCT image to one of five classes, in this fixed order: central serous chorioretinopathy (CSC), macular hole (MH), macular telangiectasia (MacTel), Stargardt disease and retinitis pigmentosa (RP). The canonical manifest holds $27/26/25/19/27$ real development images (124 in total) and $5/5/4/4/4$ legacy queries, the rare images of the existing test split (22 in total). No rare images are in the manifest's validation split; development folds provide held-out validation. The full manifest has 84,630 images: 74,082 training, 9,526 validation and 1,022 test. We will not change it; derived rosters only reference its paths. These counts are images, not verified independent patients.

Two label budgets. A five-class K-shot head sees $5K$ labelled supports. Separately, development uses 124 inspected labels in total: supports are drawn from these images, so folds and draws add no new labels. We have already inspected the 22 rare evaluation labels, so the legacy query set is not blind. Common fitting labels and upstream pretraining exposure will be reported separately. We use no synthetic supports, stochastic augmentation, self-training on unlabelled target images, query adaptation, fitted feature scaler or normalization updated from queries.

B. Dependence Screening and Feasibility Gates

Duplicate screen. Before any new-protocol score, we screen the 146 rare benchmark images in three ways: equal file hashes; equal decoded RGB pixels and dimensions, with alpha dropped for this check only; and a fixed 64-bit horizontal difference hash with Hamming distance ≤ 4. The hash uses Pillow RGB-to-L conversion, BOX resizing to 9×8 and row-major left-greater-than-right bits, with decoder and Pillow versions pinned. The screen finds candidates; it does not detect patients. A reviewer then inspects flagged raw images without encoder scores; images linked directly or through other confirmed or unresolved duplicate pairs form one component. A component with conflicting image labels is blocked until resolved.

Folds and stop rules. Rare fitting components linked to legacy queries are excluded before the fitting lock; canonical test rows do not move. Development groups join image components with verified, dataset-namespaced patient links where these exist; unknown patients stay unknown. If a verified rare patient appears under two disease labels, the selector stops for a fold amendment before any score. Each group is named by its smallest normalized POSIX member path, and class inventories are sorted case-sensitively. A fixed PCG64 permutation and round-robin assignment then give five folds. Each held-out fold needs all five classes, and the other four folds together need at least eight fitting groups per class, so each class needs at least ten eligible groups. If any check fails, the whole selector stops before scoring. We will not drop an unfavorable fold or lower K automatically.

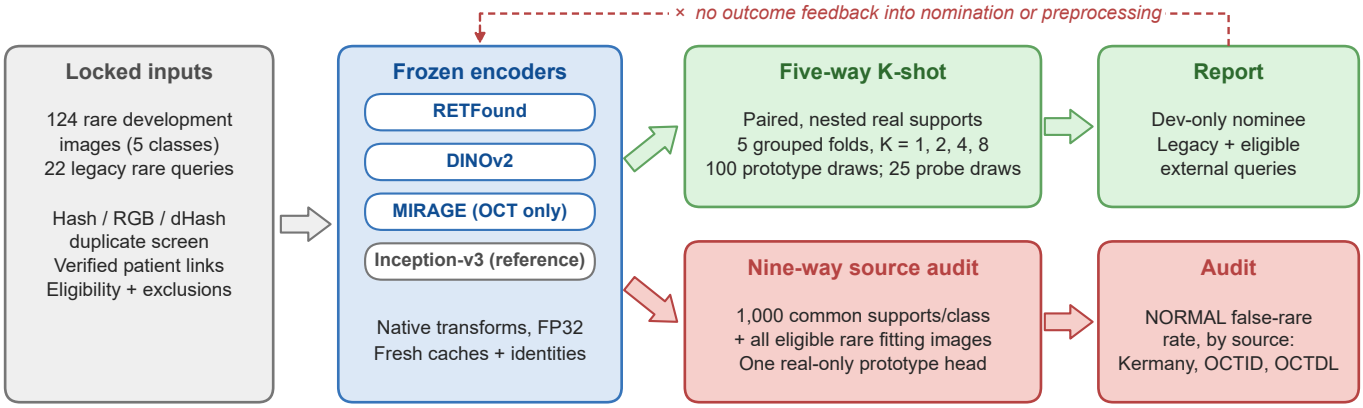


Fig. 1. Proposed protocol; all boxes denote planned operations. Shared locked image identities and frozen checkpoint pipelines feed two distinct analyses. The five-way branch uses paired K-shot supports and development-only nomination. The nine-way branch uses a separate, deterministic real-only fitting roster to measure false-rare rate (FRR) on NORMAL scans. External inventories and final query IDs are locked before their inference. Image-similarity groups do not establish patient identity. No outcome from either reporting branch feeds back into nomination or preprocessing.

TABLE II
FIXED CHECKPOINT IDENTITIES; EXACT REVISIONS AND TRANSFORMS AWAIT THE PRESCORE GATE.

Arm	Public identity / reference implementation
RETFound	`YukunZhou/RETFound_mae_natureOCT`
DINOv2	`facebook/dinov2-base`
MIRAGE	`j-morano/MIRAGE-Base`, OCT only
RETFound fallback	`microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224`, image only
Inception-v3	Historical Keras ImageNet loading/backbone; 256 → 299 resize and global average pooling

C. Frozen Checkpoint Pipelines and Feature Identity

Models. The roster is fixed: RETFound Nature OCT, DINOv2-Base and MIRAGE-Base (OCT-only inference), with an ImageNet Keras Inception-v3 reference (Table II). BioMedCLIP image features replace RETFound only if RETFound fails a predefined access, loading or terms check before any labelled comparison. If another arm or the fallback fails, the roster shrinks; we do not search for a substitute. An unavailable arm is reported as unavailable, not scored as a poor model, and an arm missing at input lock cannot enter nomination later in v1.

Feature identity. The prescore gate will fix usable terms, weight revision and hash, channel handling, deterministic native transforms, normalization, pooling, dtype and inference mode. Each checkpoint keeps its own preprocessing; crop geometry and the retained field-of-view fraction are logged per image and source. All new features, including the Keras reference, are computed in FP32 and stored as raw float32; CPU normalization and classifier arithmetic use float64. A cache's identity includes the ordered image IDs and content hashes, weights, transforms and pooling, environment, dimension and dtype, and code revision. The historical float16 Inception caches lost precision and lack this identity, so they and the saved nine-way heads stay historical artifacts. A nonfinite or zero-norm feature or centroid stops the arm.

D. Primary Prototypes and Secondary Logistic Probes

Primary classifier. Let $f_m(x) \in \mathbb{R}^{d_m}$ be encoder m's embedding, and S_c contain the K supports for class c. The primary head computes

$$u_m(x) = \frac{f_m(x)}{\|f_m(x)\|_2}, \quad a_{m,c} = \frac{1}{K} \sum_{x \in S_c} u_m(x), \tag{1}$$

$$p_{m,c} = \frac{a_{m,c}}{\|a_{m,c}\|_2}, \quad \widehat{y}(x) = \arg\max_c u_m(x)^\top p_{m,c}. \tag{2}$$

Exact ties go to the first class in CSC/MH/MacTel/Stargardt/RP order. Every support is normalized before averaging. We will not add post-hoc centering to improve an observed ranking.

Secondary probe. The logistic probe uses the same unit features and minimizes, over the $5K$ balanced supports S,

$$\mathcal{L}(W,b) = -\frac{1}{5K} \sum_{(x,y) \in S} \log[\mathrm{softmax}(W u_m(x) + b)]_y + 0.005\|W\|_F^2. \tag{3}$$

The intercept is not penalized; weights start at zero, with no class weights. We use analytic gradients and L-BFGS-B with `maxiter=1000`, `maxfun=20000`, `maxls=20`, `maxcor=10`, `gtol=1e-6` and `ftol=0`, without bounds. A fit counts as converged only if parameters and objective are finite and the largest absolute gradient is $\leq 10^{-6}$; we report actual termination reasons and missing pairs. We will not change regularization or replace draws after seeing query outcomes.

E. Paired Episodes, Nomination and Reporting

Paired episodes. Development uses $\mathcal{K}_{\mathrm{dev}} = \{1, 2, 4, 8\}$ with 100 nested support draws per fold. Each draw shuffles the fitting groups, picks one image per group uniformly at

random, and takes prefixes for larger K. Every encoder and classifier shares those IDs and within-group choices, and fold queries are all eligible held-out rows. Reporting instead uses 100 nested draws from all eligible development groups at $K \in \{1, 2, 4, 8, 16\}$, evaluated on the legacy queries and eligible external subsets. If any class has fewer than 16 fitting groups, $K = 16$ is dropped for every encoder. The secondary probe uses development $K = 4$ with the first 25 matching draws per fold, and the first 25 matching reporting draws per K. Protocol sections 4 and 6 give the exact PCG64/SeedSequence namespaces and path-order rules; we save the realized support and query IDs, not seeds alone.

Nomination. The primary metric is macro recall, the recall averaged equally over the five rare classes; ordinary accuracy, class recalls and confusion counts are secondary. Nomination uses complete prototype development results only. For model m, with $q_{f,c}$ held-out class counts and $h_{m,K,r,f,c}$ correct predictions,

$$S_m = \frac{1}{4 \cdot 100} \sum_{K \in \mathcal{K}_{\text{dev}}} \sum_{r=1}^{100} \frac{1}{5} \sum_{c \in \mathcal{C}_{\text{rare}}} \frac{\sum_{f=1}^{5} h_{m,K,r,f,c}}{\sum_{f=1}^{5} q_{f,c}}. \quad (4)$$

Pooling correct predictions within each class before averaging avoids giving folds with unequal class counts equal weight. Only FMs that complete every required fold, K and draw can be nominated, and at least two must complete. Inception is a reference, not a candidate winner. Partial averages are reported but cannot select a primary model.

Selection by score and speed. The nominee, the one FM reported as primary, is the fastest complete FM within 0.02 (two percentage points) of the best S_m, so it need not have the highest score. Timing uses the same fixed, prescore-ordered 100-image development inventory (IDs repeat only for timing if needed), batch size one, FP32, one warm pass and five timed passes on the same hardware and session. Speed is the median of the five per-pass mean warm extraction times per image; preprocessing is included, download and disk I/O excluded, with GPU synchronization. Models run sequentially in RETFound/DINOv2/MIRAGE order, one process each; hardware changes require retiming all candidates. Remaining ties follow that order, with BioMedCLIP in RETFound's slot. The two-percentage-point margin decides where to spend engineering effort; it does not mean the models are equivalent. The nominee is locked before any new legacy or external outcome, and every model is still reported. Without a valid nomination, all reporting is exploratory.

What we report. For each K, we report the paired nominee-minus-Inception macro-recall difference and all secondary model contrasts, using shared completed episodes with available and planned counts. The mean, SD and 5th/95th percentiles over support draws describe how much results vary with the choice of examples, for these images only. They are not patient confidence intervals; repeated draws add no new patients. We plan no confirmatory p-values, episode standard errors or automatic superiority or equivalence claims. Because the nominee is selected on development data, its development estimate is optimistic.

F. Separate Nine-Way Source Audit

Audit head. Each encoder gets exactly one deterministic real-only nine-way prototype head, built with (1)–(2) with K replaced by each class's support count, so averaging stays within classes. The common fitting roster is the same seed-42 sample of 1,000 images per common class, selected with `major_indices` from the full 73,958-row common fitting frame and mapped back to paths in order. All eligible real rare fitting images are added. The rare exclusion rule does not remove common rows. Before exclusions and external additions, each encoder extracts 4,000 common fitting, 1,000 common test and 146 rare images: 5,146 in total. The audit uses no common-validation images, optimizer seeds or synthetic images.

False-rare rate. Class order is fixed: NORMAL, CNV, DME, DRUSEN, CSC, MH, MacTel, Stargardt and RP. For each source s, the false-rare rate is

$$\text{FRR}_s = \frac{\#\{x \in N_s : \widehat{y}(x) \in \mathcal{C}_{\text{rare}}\}}{|N_s|}, \quad (5)$$

where $\widehat{y}$ is the nine-way prediction and N_s contains the eligible true NORMAL queries. We report Kermany, OCTID [19] and OCTDL [20] separately, with NORMAL recall, common confusions, rare recall and all denominators. A low FRR is not enough if rare recognition is poor. The five-way head has no NORMAL output, so it cannot give a meaningful FRR.

External data. Before any outcome, we freeze each external release and hash, terms, label inventory and eligibility rules. External queries are screened against every locked fitting row, the legacy queries and each other, with the same image screen and verified patient links. Fitting- or legacy-linked external components are excluded without changing locked fitting rosters; within-query dependence is recorded. Final external query IDs are frozen before inference. We state what the screen covers: overlap with unused legacy images and private pretraining data stays unknown. Rare transfer depends on verified labels, including clinical confirmation that OCTID's central serous retinopathy (CSR) is equivalent to CSC. OCTDL's broad condition labels do not by themselves identify MH. Partial-class cohorts keep all five outputs and report present-class recalls and output shares, not a five-class macro recall. All nine output shares stay visible in the source audit. Cosine scores are not called probabilities; logistic outputs include probabilities in fixed order.

Intervals. For the deterministic nine-way outputs, optional descriptive 95% percentile intervals use 5,000 patient-cluster draws (seed zero), only where verified patient IDs exist. Otherwise we label intervals as conditional on images or duplicate groups. Intervals are conditional on the fitted head and leave out development-selection uncertainty.

IV. Discussion and Execution Boundaries

Why two tests. A model may tell the rare diseases apart well and still label normal scans from a new source as rare.

A model with a low FRR may still recognize rare diseases poorly. Neither test can separate the effects of morphology, acquisition, preprocessing, model capacity or pretraining. The five-way restriction and duplicate screen address specific dependencies; they do not establish patient independence or clinical transportability.

Limitations. With four or five legacy queries per rare class, one prediction moves recall by 0.25 or 0.20. Already-inspected labels, unknown rare patients, upstream exposure and incomplete external disease coverage limit population claims. Larger K makes support sets overlap more, so a narrow spread across draws does not mean a precise population estimate.

Compute. One isolated Colab PyTorch extractor and evaluator, with Keras for Inception and CPU classifiers, will use one Colab Pro GPU session at a time. We exclude generation, augmentation sweeps, prompts, adapters and backbone fine-tuning. Runtime, peak RAM and VRAM, and download, staging, extraction, fitting and serialization costs are recorded separately; GPU-hour savings remain unmeasured.

Input lock and stop rules. Before any labelled comparison, a local input lock archives the protocol, roster, grouping, episode, checkpoint, transform, code, environment and endpoint identities. This is not public preregistration. Predictions, counts, failures and completion hashes allow identity-checked resumption. The stop rules are: (1) failed external NORMAL access leaves a rare-only pilot; (2) a failed rare mapping removes that transfer claim; (3) a numerical or identity failure stops the arm; and (4) a required-fold failure stops selection. Dated amendments keep earlier outputs.

Timeline. This 11 November submission reports the proposed methods without new FM outcomes. The planned November milestones are implementation and input checks (12–14), paired development and nomination (15–19), locked query reporting (20–23) and evidence freeze (27). Access, loading, grouping and clinical mappings are still unpassed gates. If we run late, we drop the secondary probes first and report the reduced scope; we will not wait for favorable outcomes.

V. Conclusion

Shared episodes, explicit label counts and development-only nomination make the frozen-encoder comparison checkable, and the nine-way audit tests source-dependent errors. We will report outcomes and failures without causal pretraining, patient-independent validation or clinical-benefit claims beyond the evidence.

References

[1] D. S. Kermany *et al.*, "Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning," *Cell*, vol. 172, no. 5, pp. 1122–1131.e9, 2018. [Online]. Available: https://doi.org/10.1016/j.cell.2018.02.010

[2] T. K. Yoo, J. Y. Choi, and H. K. Kim, "Feasibility study to improve deep learning in OCT diagnosis of rare retinal diseases with few-shot classification," *Medical & Biological Engineering & Computing*, vol. 59, no. 2, pp. 401–415, 2021. [Online]. Available: https://doi.org/10.1007/s11517-021-02321-1

[3] G. Quellec, M. Lamard, P.-H. Conze, P. Massin, and B. Cochener, "Automatic detection of rare pathologies in fundus photographs using few-shot learning," *Medical Image Analysis*, vol. 61, 2020, art. no. 101660. [Online]. Available: https://doi.org/10.1016/j.media.2020.101660

[4] Y. Zhou *et al.*, "A foundation model for generalizable disease detection from retinal images," *Nature*, vol. 622, no. 7981, pp. 156–163, 2023. [Online]. Available: https://doi.org/10.1038/s41586-023-06555-x

[5] J. Morano *et al.*, "Multimodal foundation model and benchmark for comprehensive retinal OCT image analysis," *npj Digital Medicine*, vol. 8, no. 1, 2025, art. no. 576. [Online]. Available: https://doi.org/10.1038/s41746-025-01852-3

[6] D. Shi *et al.*, "A multimodal visual–language foundation model for computational ophthalmology," *npj Digital Medicine*, vol. 8, no. 1, 2025, art. no. 381. [Online]. Available: https://doi.org/10.1038/s41746-025-01772-2

[7] M. Oquab *et al.*, "DINOv2: Learning Robust Visual Features without Supervision," *Transactions on Machine Learning Research*, 2024. [Online]. Available: https://openreview.net/forum?id=a68SUt6zFt

[8] Y. Zhou *et al.*, "Generalist versus Specialist Vision Foundation Models for Ocular Disease and Oculomics," arXiv preprint arXiv:2509.03421, 2025, version 1, submitted September 3, 2025. [Online]. Available: https://arxiv.org/abs/2509.03421v1

[9] R. Chen *et al.*, "Boosting foundation models for rare eye disease diagnosis via a multimodal text-to-image generative framework," *npj Digital Medicine*, vol. 9, no. 1, 2026, art. no. 371. [Online]. Available: https://doi.org/10.1038/s41746-026-02560-2

[10] J. Snell, K. Swersky, and R. Zemel, "Prototypical Networks for Few-shot Learning," in *Advances in Neural Information Processing Systems*, I. Guyon *et al.*, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/cb8da6767461f2812ae4290eac7cbc42-Paper.pdf

[11] Y. Wang, W.-L. Chao, K. Q. Weinberger, and L. van der Maaten, "SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning," *arXiv preprint arXiv:1911.04623*, 2019.

[12] Y. Guo *et al.*, "A Broader Study of Cross-Domain Few-Shot Learning," in *Computer Vision – ECCV 2020*, 2020, pp. 124–141.

[13] C. P. Phoo and B. Hariharan, "Self-training for Few-shot Transfer Across Extreme Task Differences," in *International Conference on Learning Representations*, 2021. [Online]. Available: https://arxiv.org/abs/2010.07734

[14] M. Mosbach, T. Pimentel, S. Ravfogel, D. Klakow, and Y. Elazar, "Few-shot fine-tuning vs. in-context learning: A fair comparison and evaluation," in *Findings of the Association for Computational Linguistics: ACL 2023*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds. Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 12 284–12 314. [Online]. Available: https://aclanthology.org/2023.findings-acl.779/

[15] J. Silva-Rodríguez, S. Hajimiri, I. Ben Ayed, and J. Dolz, "A Closer Look at the Few-Shot Adaptation of Large Vision-Language Models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024, pp. 23 681–23 690.

[16] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Conditional Prompt Learning for Vision-Language Models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 16 816–16 825.

[17] D. He *et al.*, "Boosting pathology foundation models via few-shot prompt-tuning for rare cancer subtyping," *Nature Communications*, vol. 17, no. 1, 2026, art. no. 5058. [Online]. Available: https://doi.org/10.1038/s41467-026-71715-2

[18] C. Lian, H.-Y. Zhou, Z. Hu, and J. Qin, "BenchReAD: A Systematic Benchmark for Retinal Anomaly Detection," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, ser. Lecture Notes in Computer Science, vol. 15961. Springer Nature Switzerland, 2025, pp. 35–45. [Online]. Available: https://doi.org/10.1007/978-3-032-04937-7_4

[19] P. Gholami, P. Roy, M. K. Parthasarathy, and V. Lakshminarayanan, "OCTID: Optical coherence tomography image database," *Computers & Electrical Engineering*, vol. 81, 2020, art. no. 106532. [Online]. Available: https://doi.org/10.1016/j.compeleceng.2019.106532

[20] M. Kulyabin *et al.*, "OCTDL: Optical Coherence Tomography Dataset for Image-Based Deep Learning Methods," *Scientific Data*, vol. 11, no. 1, 2024, art. no. 365. [Online]. Available: https://doi.org/10.1038/s41597-024-03182-7